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Technologies

The Semiconductor Foundry Industry

Landscape and Competitive Analysis

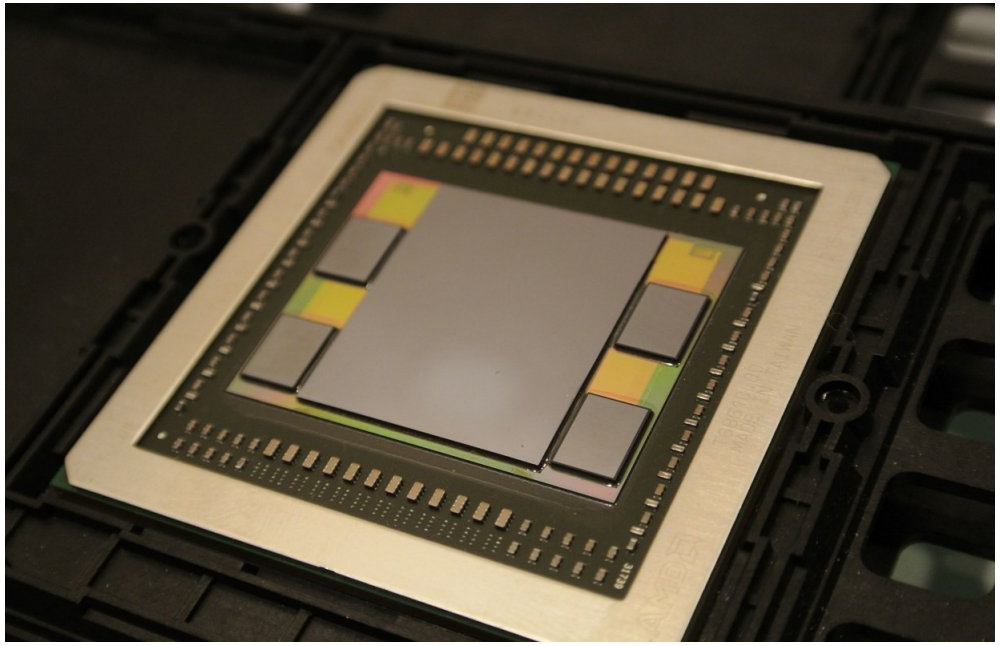
Ecosystem Team | Ziqi Wang | August 2026

Part II of the Semiconductor Ecosystem series — follows “Chip Fabrication Process”

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Where the foundry sits in the chip supply chain

Nine steps to build an AI accelerator. Two are single-sourced, and both are the same company.



One logic die, four HBM stacks, one interposer — an H100 is built the same way (AMD Fiji shown)

Instruction set / IP	Arm, RISC-V, Synopsys, Cadence, Rambus
Design & architecture	NVIDIA
EDA & verification	Synopsys, Cadence, Siemens EDA
Photomasks	TSMC in-house, Toppan, Photronics
Logic wafer (N4)	**TSMC — no qualified alternative**
HBM3 memory	SK hynix, Micron, Samsung
Advanced packaging (CoWoS)	**TSMC — plus a thin OSAT tail**
Test & burn-in	ASE, Amkor, KYEC
Substrate	Ibiden, Shinko, Unimicron

In 1995, twenty-six firms could build the leading edge. Today **three** can, and **one** takes **72%** of foundry revenue.

How it converged, the map today, what to watch.

Source: TrendForce Q1 2026 foundry ranking; NVIDIA H100 brief. Photo: Wikimedia.

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2026

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10 pp

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Four variables you can actually check against reality

3 pp

Design note:
History earns 12 pages because it establishes three words — **trust**, **capital**, **lock-in** — that every later argument points back to.

Source: author's structure.

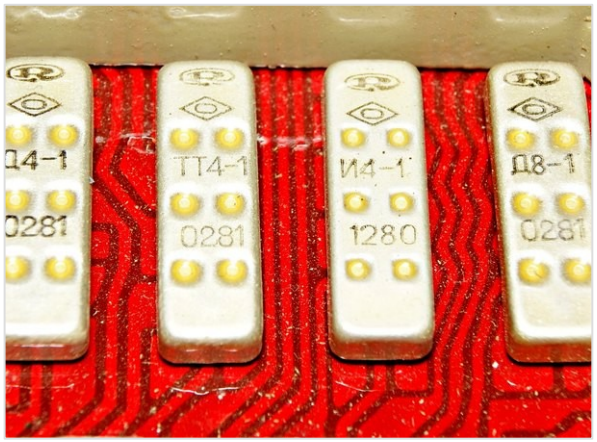
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The IDM era: design and manufacturing in one company

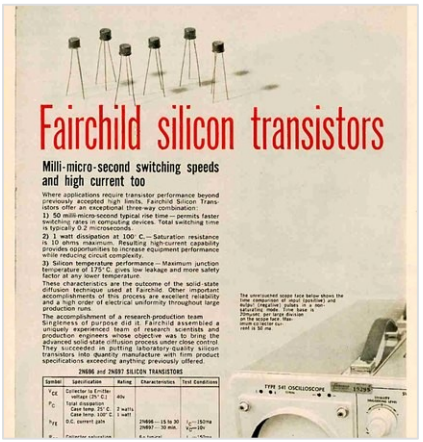
> **A technical necessity, not a business preference: there was no portable interface between design and process.**



Fairchild, Mountain View: the first commercially practicable IC, 1959



Early germanium integrated circuits



Fairchild trade advertisement, 1958



Robert Noyce and Gordon Moore, Intel, 1970

No portable design rules

Written against one line's lithography, furnaces and metal stack. Move it and it broke.

The process defined the product

Design and process were one conversation. Splitting them lost the yield learning in between.

So only IDMs could exist

TI, Fairchild, Motorola, Intel, National, RCA, NEC, Toshiba, Philips — all owned wafers, masks and test.

Takeaway — the division of labour was not inevitable. It required two separate enablers.

Source: Computer History Museum; Fairchild Semiconductor archives. Photos: Wikimedia Commons.

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Enabler 1: portable design rules

Mead–Conway made layout portable; MOSIS made a wafer a shared resource.



Carver Mead, Caltech



Lynn Conway, Xerox PARC

1978–1980 · Introduction to VLSI Systems

Mead and Conway replaced process-specific craft with a **scalable λ-based rule set**. A layout could be re-targeted to any line that met the rules — a **document**, not a recipe.

1981 · MOSIS and the multi-project wafer

DARPA funded a broker that **aggregated many small designs onto one wafer**, run by an outside fab. Working silicon for a few thousand dollars: design without a fab, proven a decade early.

The abstraction stack this created

- Application / system
- RTL and logic
- Cell library and IP
- **Layout in portable design rules**
- Process and equipment

The cut runs at the red layer.

Above it: sold as a design.
Below it: sold as a service.

That seam is the foundry industry.

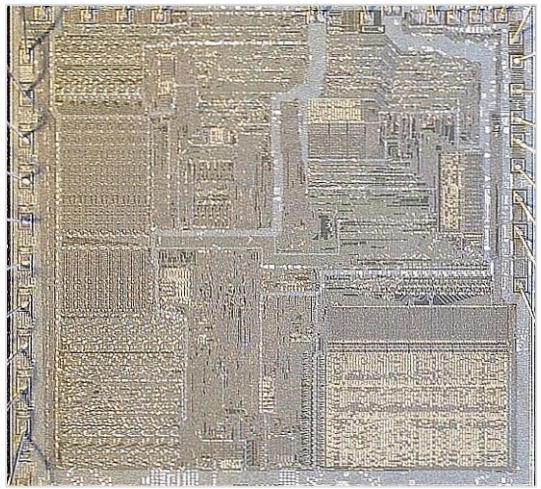
Still missing: a manufacturer with no competing interest. MOSIS bought spare capacity from IDMs — companies that sold competing chips.

Source: Mead & Conway, Introduction to VLSI Systems (1980); MOSIS/USC ISI programme history.

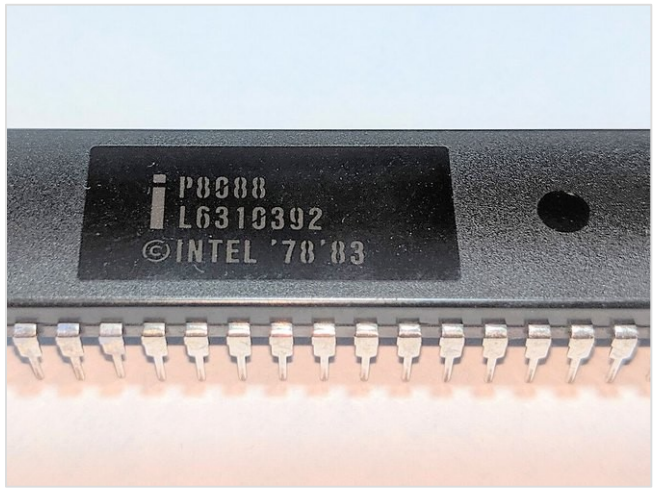
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Enabler 2: second sourcing, and its limit

> **Transferring a design to another manufacturer worked commercially, but only under customer pressure.**



Intel 8086 die



8088 in DIP-40 — IBM's required second source

1976–1982 · the second-source agreement

Buyers refused single-supplier parts. To win the IBM PC, Intel **licensed AMD** to build the 8086/8088. Motorola–Hitachi, TI–AMD and Zilog–Mostek followed.

What it proved

A design could be built by a company that did not create it. The handoff worked.

What it did not solve

Your second source was your **competitor**. Intel and AMD litigated for two decades. Nobody hands a roadmap to a rival.

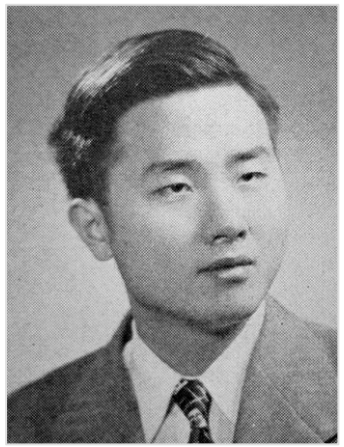
The two enablers, and what was still missing

SOLVED 1981	**Technical**	Portable design rules plus shared multi-project wafers
SOLVED 1982	**Commercial**	Proof that an outside party can build your design
MISSING	**Structural**	A manufacturer that will never compete with you

Source: Intel/AMD 1982 technology exchange agreement; IBM PC sourcing requirements.

TSMC, 1987: the first pure-play foundry

> **TSMC was founded to manufacture chips designed by others, with no products of its own.**



Morris Chang, MIT, 1951



Founder, TSMC



Hsinchu Science Park — the industrial policy that hosted it

The founding position

Passed over for CEO at TI, Chang moved to Taiwan at 54 to head ITRI. In 1987 he spun a pilot line into a company with no products.

Who paid for it

Philips 27.5% — process licence and patent cover.

Taiwan government ≈48% via the Development Fund.

Private capital ≈25%, much of it reluctant.

Intel, TI and others were asked to invest. All declined.

Why the industry thought it would fail

“A fab with no products is a fab with no floor.”

Utilisation was assumed to be uncontrollable without an internal product line.

“Nobody will trust their designs to a stranger.”

Second-source history said the manufacturer eventually becomes a rival.

“The leading edge belongs to whoever also designs.”

Process–design co-optimisation was believed to require one roof.

Source: TSMC corporate history; Philips–TSMC 1987 joint venture terms; ITRI records.

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The pure-play business model

> **“We don’t design chips,
> so we never compete with our customers.”**
>
> — Morris Chang, on why a pure-play foundry can exist at all



A foundry does not sell capacity. It sells a credible promise not to become a competitor.

What the customer is really buying

An assurance that what they hand over never reappears in a rival product.

Why this cannot be copied by an IDM

Samsung sells memory and phones; Intel sells CPUs — and learns from every design win.

Why it compounds

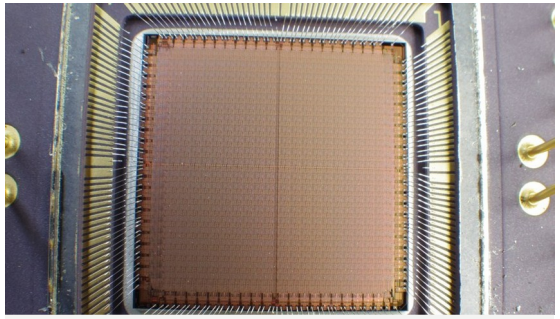
Neutrality → customers → wafers → yield learning → price → more customers.

Source: TSMC investor communications; Morris Chang public remarks. Photo: Wikimedia Commons.

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Fabless design: the other half of the split

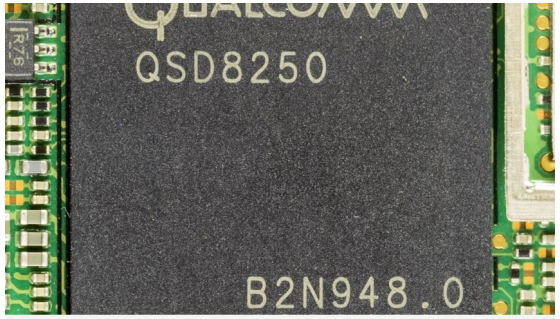
> A foundry is only viable if fabless companies exist, and they could only exist once it did.



Xilinx

1984

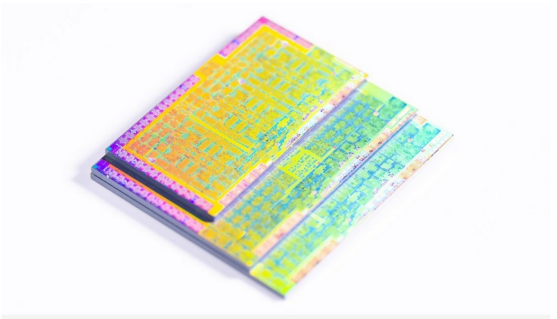
First fabless FPGA company; every wafer outsourced from day one



Qualcomm

1985

Modem IP plus outsourced silicon became the mobile standard



NVIDIA

1993

Founded assuming TSMC existed; today the largest single consumer of leading-edge



Broadcom

1991

Connectivity and custom ASIC; now the anchor of custom AI silicon

Two industries, one enabling event

Xilinx (1984) and Qualcomm (1985) predate TSMC, living on IDM spare capacity — allocated last, repriced at will. After 1987 a business plan no longer needed a fab.

The dependency runs both ways: without fabless customers no pure-play fab fills. Neither half works alone, which is why the split held.

Source: Company founding records; SIA fabless revenue series. Photos: Wikimedia Commons.

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The followers: structure copied, position not

> **Within thirteen years everyone had a pure-play foundry. Almost none are still at the leading edge.**



UMC, Hsinchu



SMIC, Shenzhen



GlobalFoundries, Dresden (ex-AMD Fab 36)

The copies

Year	Company	Base	Outcome
1987	**Chartered Semiconductor**	Singapore	State-backed pure-play; sold to GlobalFoundries in 2010
1995	**UMC converts to pure-play**	Taiwan	Spun out its own design teams to remove the conflict; stopped investing past 14nm
2000	**SMIC**	China	Pure-play from the start; now capped by equipment export controls
2009	**GlobalFoundries**	US / Germany	AMD's fabs, recapitalised by Abu Dhabi; abandoned 7nm in 2018

The structure was public from day one. What could not be copied: fifteen years of yield learning, a customer base big enough to fund the next node, and the discipline to keep going.

Source: Company filings and corporate histories. Photos: Wikimedia Commons.

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Why owning a fab stopped making sense

> **“Real men have fabs.”**
>
> — Jerry Sanders, founder and CEO of AMD, 1990s
> AMD spun off its fabs in 2009. Sanders had retired in 2002.



Jerry Sanders, AMD



The asset in question

Not a joke about machismo. At the time it was an accurate description of the economics.

Why it was true until ~2000

A fab cost $1–2B. A company with $3B of revenue could fund one — and owning it bought co-optimisation nobody could buy.

What broke it

Each node doubled the tool cost while shrinking the customer list. Fixed cost stopped scaling with any one company's revenue.

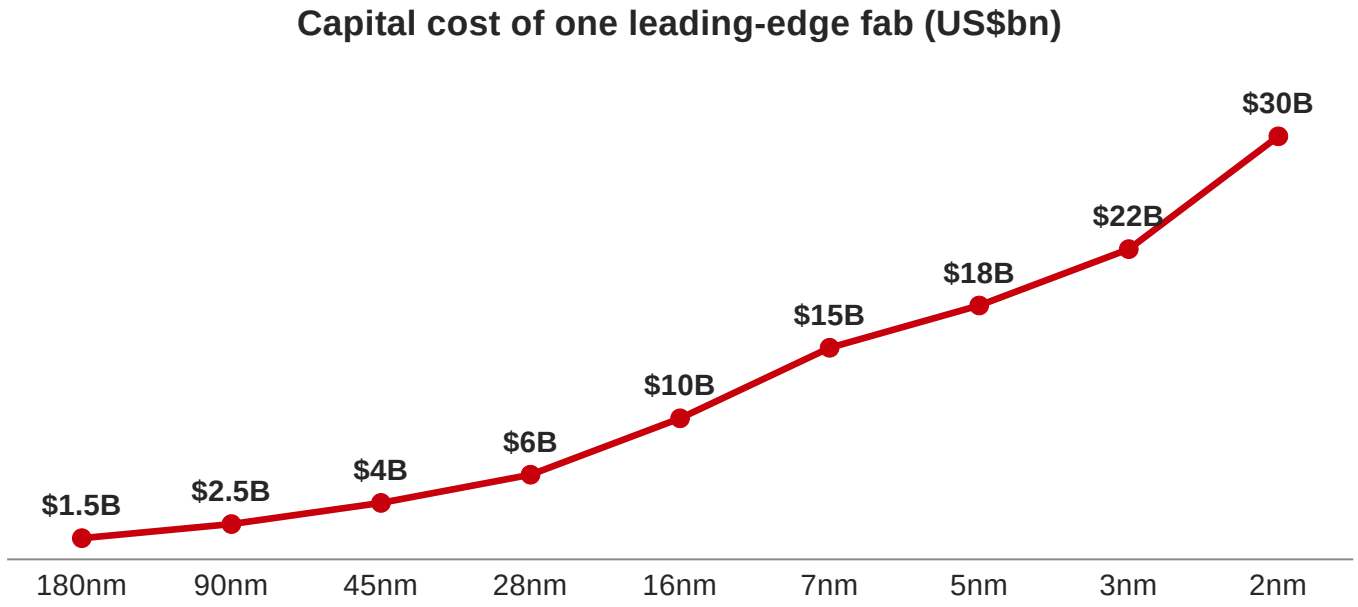
What it became

By 2018 a fab cost $15–20B and needed 40,000+ wafers a month. Only a company aggregating everyone's volume could fill one.

Source: Attributed to Jerry Sanders, AMD; AMD 2008–09 asset-smart transaction filings.

Cost, not physics, became the barrier

The constraint is not whether a company can build the node, but whether it can fund design and capacity first.



Capital cost of one leading-edge fab, 1999–2026 — indicative

The second cheque: design

28nm **$50–90M**
Still affordable to a mid-size fabless company

7nm **$250–300M**
Viable above roughly $1B of product revenue

5nm **$450–540M**
The customer list is now a short list

2nm **≥ $725M**
Fewer than 20 companies can justify a tape-out

A wafer at N2 now costs roughly $30,000, against $3,000 at 28nm.

Both cheques clear before the first revenue wafer ships. That filter removed twenty-three companies.

Source: IBS design-cost estimates; Silicon Analysts 2026 wafer pricing; company capex. Indicative.

Exits from the leading edge, 2008–2019

Each exit below was a rational decision by a competent company, which is what makes it structural.

2008–09	**AMD**	Spins manufacturing into GlobalFoundries with Abu Dhabi capital; AMD goes fabless.
2014–15	**IBM**	Pays GlobalFoundries US$1.5 billion to take its fabs away — a negative price for capacity.
2018	**GlobalFoundries**	Abandons 7nm outright. Redeploys to mature and RF/FDX platforms — and becomes profitable doing it.
2018	**UMC**	Stops advanced-node investment at 14nm. Same conclusion, reached publicly and calmly.
2015–19	**Intel**	10nm slips repeatedly. Thirty years of leadership lost in one node transition, not yet recovered.

What none of them lacked

Not talent, process IP or patents. What they lacked was external volume to amortise the next node.

Source: AMD 2009 8-K; IBM–GlobalFoundries 2015; GlobalFoundries 2018; UMC 2018 statements.

The three survivors and their economics

None survived by being best at lithography. Each solved amortisation a different way.

TSMC

Aggregation + neutrality

MECHANISM
Pools the whole fabless industry into one fab fleet: the node amortises across hundreds of customers, not one product line.

COST OF THE STRATEGY
Must stay out of chip design permanently, including the most profitable products it manufactures.

WHERE IT IS FRAGILE
Geographic concentration, and a customer base that would fund an alternative if one became credible.

Samsung

Internal demand floor

MECHANISM
Memory, mobile SoC and image sensors guarantee a base load, so the fab stays full without winning every external deal.

COST OF THE STRATEGY
Sells products that compete with its own customers — the problem the pure-play model was invented to remove.

WHERE IT IS FRAGILE
External customers structurally discount it; yield reputation compounds the discount.

Intel

Product margin subsidy

MECHANISM
x86 server and client margins funded process development for three decades without any external customer.

COST OF THE STRATEGY
Only as durable as the product franchise — now contested by Arm and by custom silicon.

WHERE IT IS FRAGILE
The hardest transition in the industry: becoming neutral while still selling competing products.

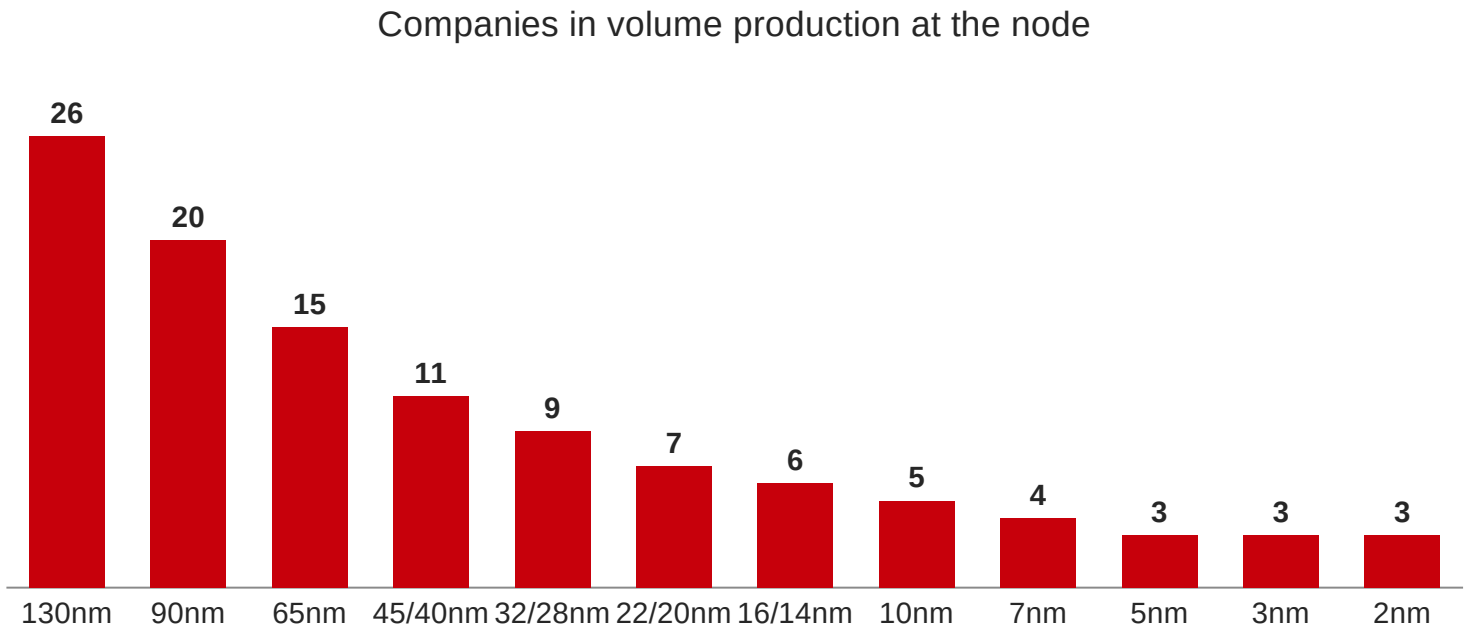
Part 5 tests each of these three mechanisms against 2026 conditions.

Source: Company filings and disclosed strategy; author's framing.

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Companies able to build each node

The chart from the fabrication deck, extended forward.



Who is left at 2nm

TSMC
N2 · nanosheet GAA
HVM since Dec 2025

Intel
18A · RibbonFET + PowerVia
HVM since late 2025

Samsung
SF2 · GAA (2nd gen)
Ramping; yield contested

Rapidus
2nm GAA
Pilot line; HVM target 2027

Three companies — and the third has no external customer at scale. From here: what happens when the leading edge is a duopoly with an asterisk, and the mature edge is not.

Source: Company disclosures and press reports; node counts are approximate and count volume production only.

Three independent ways to segment foundries

A foundry cannot be ranked on nanometres alone. Mixing the axes is the most common error.

AXIS 1

BY NODE

Leading ≤3nm
Advanced 5–16nm
Mainstream 22–65nm
Mature ≥90nm

Use it to talk about technology competition

AXIS 2

BY BUSINESS MODEL

Pure-play
IDM turned foundry
Specialty foundry
National champion

Use it to talk about strategic motive

AXIS 3

BY PROCESS TYPE

Logic · BCD power
RF-SOI · CIS · eNVM
GaN / SiC · MEMS
Analog / HV

Use it to talk about end markets

Why keeping them separate matters

GlobalFoundries	Mainstream and mature on axis 1, pure-play on 2, RF-SOI and FDX on 3. By nanometres a loser; by margin, not.
Samsung Foundry	Leading edge on axis 1, IDM-turned-foundry on 2. Its problem is entirely axis 2.
Hua Hong	Mature on 1, national champion on 2, specialty analog and power on 3. Capacity answers to policy, not price.

Source: author's framing; used consistently in Parts 4 and 5.

Axis 1 — by node

Each layer has a different player list, a different pricing mechanism and a different reason to invest.

LAYER	NODE RANGE	WHO IS THERE	WHAT IT MAKES
LEADING	**≤ 3nm**	TSMC · Samsung · Intel Foundry (+ Rapidus in pilot)	AI accelerators, flagship mobile SoC, high-end server CPU
Sold out before it exists. Price set by scarcity, not cost.			
ADVANCED	**5 – 16nm**	TSMC · Samsung · Intel · SMIC (7nm, DUV multi-patterning)	Mainstream mobile, networking, GPU, automotive compute
Still capital-intensive; the last layer where a newcomer can plausibly enter.			
MAINSTREAM	**22 – 65nm**	TSMC · UMC · GF · SMIC · Hua Hong · Nexchip · PSMC · Tower	MCU, PMIC, display driver, image sensor, connectivity, eNVM
Genuinely competitive. Chinese capacity growth sets the price ceiling.			
MATURE	**≥ 90nm (incl. 200mm)**	VIS · Tower · X-FAB · DB HiTek · SkyWater · Hua Hong · PSMC	Analog, power/BCD, high-voltage, RF, MEMS, discrete
Wins on process recipes and long-lived qualifications, not on density.			

The player list widens as you go down, and so do the ways to win. Only the top layer is a three-horse race priced by scarcity.

Source: Company node roadmaps and capacity disclosures, 2026.

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Axis 2 — by business model

Two companies at the same node price, allocate and invest differently if their reason for being there differs.

PURE-PLAY

TSMC · UMC · GlobalFoundries · SMIC · VIS · PSMC · Nexchip

Motive: fill the fab, protect neutrality.

Behaviour: allocates to whoever pays; no product markets; capex tracks booked demand.

IDM TURNED FOUNDRY

Samsung Foundry · Intel Foundry · Texas Instruments (limited) · Bosch

Motive: monetise a fab that exists for internal products.

Behaviour: external customers rank behind internal ones when capacity is tight — exactly when it matters.

SPECIALTY FOUNDRY

Tower · X-FAB · DB HiTek · SkyWater · Vanguard · Silterra

Motive: own a process recipe nobody else supports.

Behaviour: competes on qualification depth and 10–15 year supply commitments, not on nanometres.

NATIONAL CHAMPION

SMIC · Hua Hong · Nexchip · Rapidus · Tata Electronics

Motive: sovereign supply, subsidised.

Behaviour: builds capacity ahead of demand and tolerates returns a listed competitor could not.

This axis, not the node axis, explains why Samsung cannot close the gap with technology alone, and why Chinese mature capacity does not behave like a rational competitor.

Source: Author's framing; company strategy disclosures.

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Axis 3 — by process type

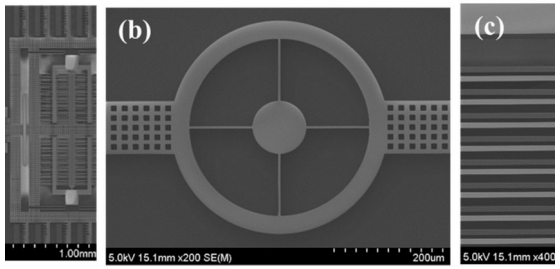
Below the leading edge, competition is about supported process recipes, not density.



CMOS image sensor

Sony, Samsung, SMIC, Hua Hong, Tower

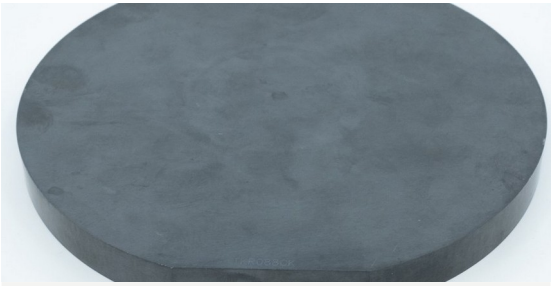
Stacked pixel + logic; wafer bonding is the differentiator



MEMS & sensors

X-FAB, Silex, Tower, TSMC (specialty)

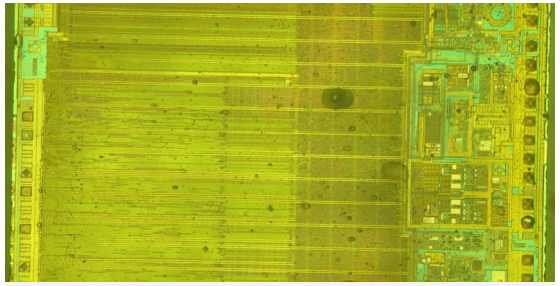
Every device is a custom process; qualification takes years



GaN / SiC power

Wolfspeed, X-FAB, TSMC, Bosch, Infineon

Substrate-limited, not lithography-limited



RF, connectivity, MCU

GF (RF-SOI), UMC, TSMC, Nexchip, PSMC

22–65nm bulk and FD-SOI; volume lives here, not at 3nm

Why specialty is defensible

An automotive BCD process qualified in 2019 must be supplied into the mid-2030s. Re-qualifying is expensive, and a newer node does not make a better 40V LDMOS. Process stability, not density.

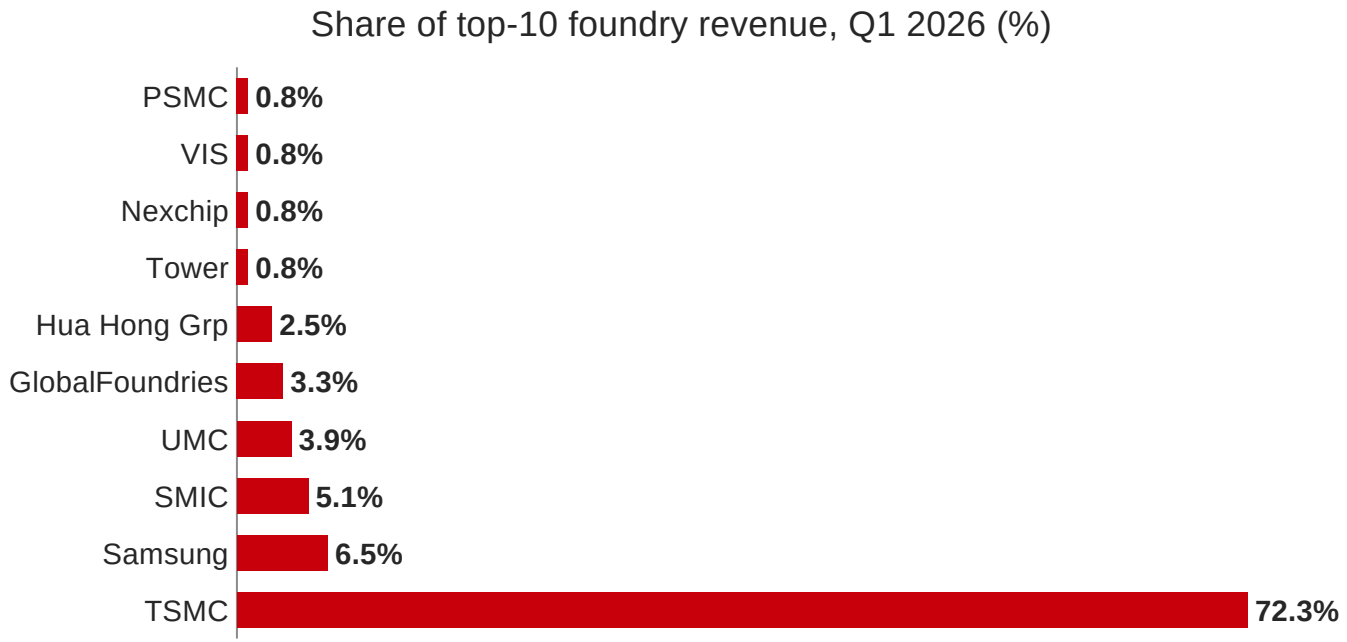
And why it is not a hiding place either

Margins are good, but the market grows with cars, not AI — and Chinese specialty capacity is expanding into these same processes. Defensible, not expanding.

Source: Company process portfolios; TrendForce specialty capacity tracking. Photos: Wikimedia Commons.

Foundry revenue and share, Q1 2026

Q1 2026 was a record quarter for the industry, and TSMC gained share within it.



Top-10 foundry revenue reached a record US$47.95bn in Q1 2026, up 3.7% QoQ

What the quarter actually said

TSMC — **US$35.86bn, +6.3% QoQ**

Gained share in what is normally a seasonal trough, on AI HPC demand and tariff-driven pre-building.

Samsung — **US$3.2bn, −5.8% QoQ**

Fell to 6.5%. TSMC is now roughly eleven times its size in foundry revenue.

SMIC — **US$2.51bn**

Within US$700m of Samsung. The number two slot is genuinely contested.

The tail — **Four players at 0.8% each**

Tower, Nexchip, VIS and PSMC are indistinguishable at this resolution — which is itself the finding.

Before quoting any of these numbers, read the next page: there are at least three definitions of “foundry”.

Source: TrendForce, Q1 2026 global top-10 foundry revenue ranking (published June 2026).

Market share depends on the definition used

Foundry market share depends on which denominator is used.

TOP-10 PURE-PLAY+

72%

TrendForce. Denominator: the ten largest dedicated foundry businesses, Samsung and Intel foundry included.

Best for: comparing foundries with each other.

TOTAL FOUNDRY MARKET

≈38%

All wafer foundry revenue worldwide (~US$320bn in 2025, +16% YoY), including the small and captive tail.

Best for: sizing the industry.

FOUNDRY 2.0

≈28–33%

TSMC's 2024 redefinition. Adds packaging, test, mask-making and non-memory IDM work — a US$250bn+ pool.

Describing the real competitive set. Also the number that looks least like a monopoly.

Why TSMC proposed Foundry 2.0 in July 2024

Two honest reasons and one convenient one: IDMs do sell foundry services, packaging is part of the same decision — and 28% attracts less antitrust attention than 72%.

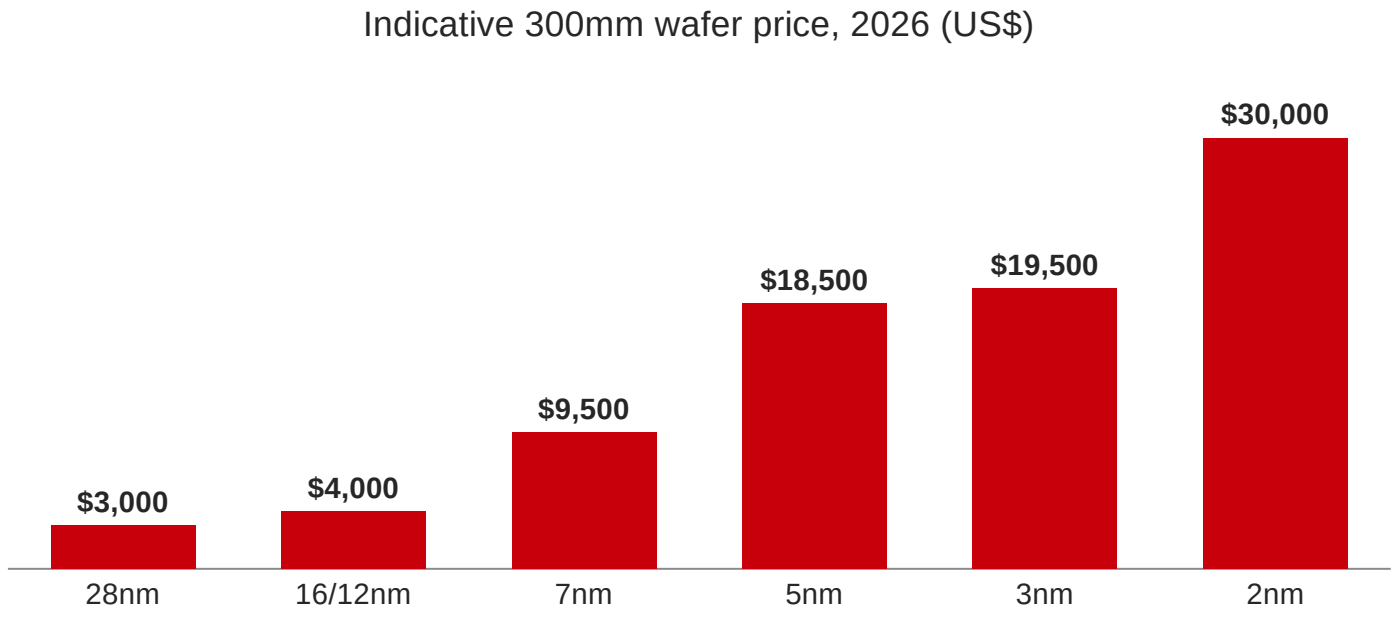
When someone quotes a foundry share, ask which denominator. Most disagreements about TSMC's dominance are about scope, not facts.

Source: TrendForce; TSMC Foundry 2.0 definition (July 2024); Tom's Hardware 2025 foundry market total.

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Wafer price by node

A 2nm wafer costs about ten times a 28nm wafer.



Indicative TSMC list pricing; large customers negotiate 15–30% below these levels

What wafer price determines

It selects customers

At US$30,000 a wafer plus US$725m of design, N2 needs roughly US$2–3bn of expected revenue.

It rewards die size discipline

Price per wafer is fixed; price per good die is not. Yield and reticle use decide affordability.

It funds the next node

Leading-edge margin pays for the fab after next. Which is why TSMC defends price, not share.

Samsung discounts into it

Typically 15–30% below TSMC — and still short of clearing the trust discount.

Page 15 shows the same narrowing on the supply side: three companies can build the node, and fewer can afford to buy it.

Source: Silicon Analysts 2026 wafer pricing series; IBS design-cost estimates. Figures indicative, not quotations.

The capital model: depreciation drives cost

Capital intensity and depreciation, not wafer processing, set the cost structure.



A wafer track line — the asset being depreciated

TSMC capex 2026	**US$52–64bn**
Raised twice this year; 70–80% to advanced nodes, the rest to packaging, masks and specialty.	
Capex intensity	**~40–45% of revenue**
Sustained for years. No other manufacturing industry at this scale reinvests at this rate.	
Depreciation	**Largest single line in COGS**
Tools are written down over roughly five years, which is shorter than the useful life of the node.	
Samsung / Intel	**Comparable absolute capex**
But spread across memory or products — a different burden per wafer.	

Three consequences that fall straight out of the model

Utilisation is the P&L

Depreciation does not care whether the fab runs. Every point of utilisation drops straight to margin.

Capacity leads demand by ~2 years

The fab you need in 2028 has to be committed now, against a forecast that cannot be right.

Scale is self-reinforcing

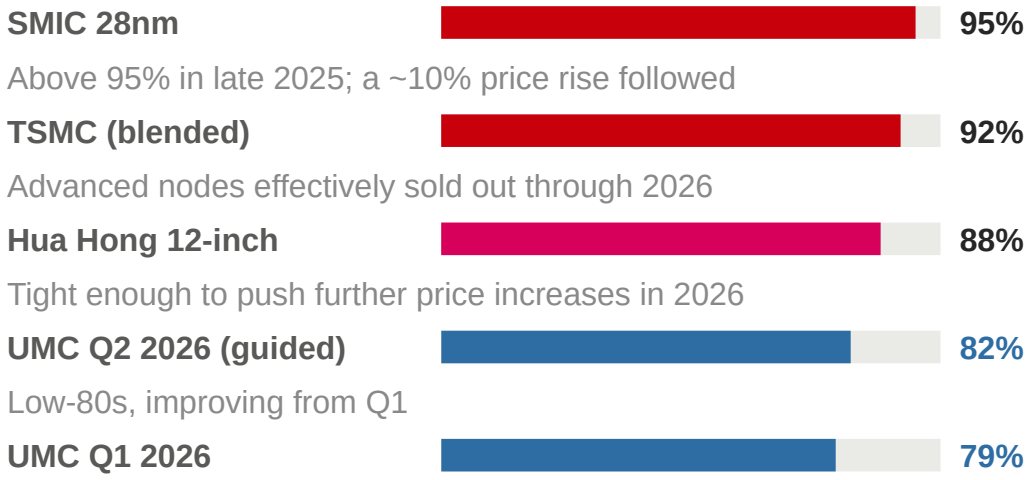
More wafers spread the same fixed cost → better price → more wafers. The lead widens by default.

Source: TrendForce and company capex guidance, 2026; TSMC quarterly disclosures. Photo: Wikimedia Commons.

Utilisation and yield

Price is negotiated once a year. Utilisation and yield move every week, and move margin further.

Utilisation — where the industry actually sits



Rule of thumb: ±10 points of utilisation ≈ ±5–7 points of gross margin.

Yield — the one barrier that is still technical

TSMC N2 defect yield is around 65% today, ~75% expected at maturity. Intel 18A and Samsung SF2 are believed behind; neither publishes.

Why yield, not lithography, is the barrier

Everyone buys the same EUV scanners from the same supplier. What they cannot buy is the defect-density learning curve — a function of cumulative wafers run. Most customers means fastest learning.

Volume buys yield, yield buys margin, margin buys the next node, the next node buys volume. Nobody has broken in from outside since 1987.

Source: TrendForce; UMC Q1 2026 call; supply-chain reports on N2 yield. Utilisation approximate.

Why customers do not switch foundries

Switching cost, not process technology, is what keeps customers in place.

PDK maturity

A PDK encodes years of silicon-correlated models. A young one means more margin, more respins, later tape-out.

IP availability

SerDes, PHY, PLL, SRAM compilers — silicon-proven per foundry, per node. Missing IP is schedule risk.

EDA certification

Synopsys and Cadence flows are certified per foundry, per node. An uncertified combination is nobody's to own.

Shuttle / MPW cadence

Frequent shuttles de-risk a node cheaply. A thin calendar means committing to a mask set on faith.

Requalification

Automotive and industrial parts need AEC-Q100 requalification: 12–24 months before first revenue.

Learning and staff

Your physical design team's intuition is foundry-specific. Switching resets it.

What this costs in practice

A serious second-source qualification on a complex SoC runs 12–24 months and tens of millions before one revenue wafer. Apple, NVIDIA and Qualcomm have funded dual-track work — but not for a 10% price difference.

So matching PPA is not enough. A challenger must be better for long enough to justify a two-year requalification — while the incumbent ships the next node.

Source: Foundry PDK/IP ecosystem documentation; customer qualification practice.

Six areas of competition in 2026

Each company sits differently in each area. A single ranking hides that.

1 THE 2nm GENERATION

Nanosheet GAA vs RibbonFET plus backside power. The fork is power delivery, not the transistor.

pp 27–28

2 ADVANCED PACKAGING

CoWoS/SoIC vs Foveros/EMIB-T vs SAINT. The AI bottleneck moved from the front end to the back.

pp 29–30

3 MATURE-NODE ECONOMICS

Chinese capacity, TSMC's retreat from 28nm, and a 2026 rebound more fragile than it looks.

p 31

4 SPECIALTY PROCESS

Automotive, industrial, power, RF — the only ground where the second tier does not meet TSMC.

p 32

5 GEOPOLITICS & SUBSIDY

Section 232 tariffs, an expiring tax credit, export controls, three overseas fabs behind schedule.

p 33

6 ECOSYSTEM LOCK-IN

EDA certification, IP libraries, PDK maturity, shuttle cadence — nobody is close.

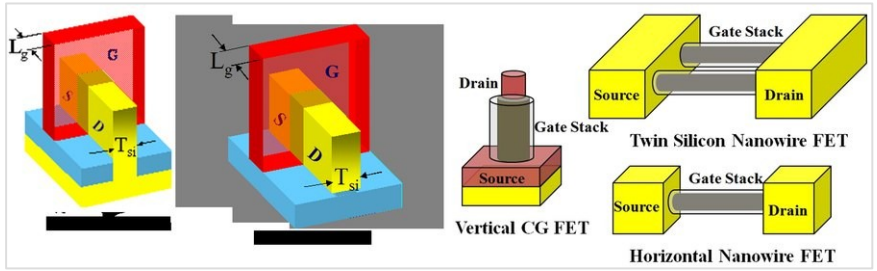
p 34

Areas 1–2 decide who leads, 3–4 who survives, 5–6 how fast either changes.

Source: Author's framing.

Area 1 — 2nm: the difference is power delivery

All four contenders moved to gate-all-around. What separates them is where the power rails run.



FinFET to nanosheet / gate-all-around device structures

What backside power actually buys

Moving power delivery to the back of the wafer frees front-side metal for signal routing and cuts IR drop — about a node's worth of performance-per-watt.

Intel shipped it first. Page 28 is why that has not converted into customers.

TSMC

N2

Nanosheet GAA

HVM since Dec 2025

BACKSIDE POWER
Not in N2 — arrives with A16

YIELD
~65% today, ~75% at maturity

POSITION
Booked out; ~20% more capacity in 2026

Intel

18A

RibbonFET + PowerVia

HVM since late 2025

BACKSIDE POWER
PowerVia shipped first — a real lead

YIELD
Believed behind N2; not published

POSITION
Volume is Intel's own Panther Lake

Samsung

SF2

2nd-gen GAA (MBCFET)

Ramping

BACKSIDE POWER
Planned for SF2Z, later

YIELD
The persistent question; 15–30% discount

POSITION
Needs one flagship external win

Rapidus

2nm

GAA, IBM-derived

Pilot line since 2025

BACKSIDE POWER
Roadmap item

YIELD
Prototypes demonstrated Jul 2025

POSITION
HVM target H2 2027; state funded

Source: company disclosures; SemiWiki and TrendForce, 2026. Yields are supply-chain estimates.

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Area 1 continued — customer commitments

Intel has the stronger power-delivery position; TSMC has the customers.

Where the leading-edge demand actually sits

Apple	**TSMC N2**	First and largest N2 allocation
NVIDIA	**TSMC N3 / N2**	Plus effectively all of CoWoS
AMD	**TSMC N2**	Venice server silicon
Qualcomm / MediaTek	**TSMC N2**	Both evaluated Samsung; both stayed
Broadcom / Google / Amazon	**TSMC**	Custom AI silicon — the fastest-growing pool
Intel (internal)	**Intel 18A**	Panther Lake; captive volume
Tesla	**Intel 14A**	Confirmed external commitment
Samsung (internal)	**Samsung SF2**	Exynos; captive volume

Intel Foundry: the state of play in August 2026

18A is in volume production, but the volume is Intel's own. The pitch has moved to 14A — first external commitments targeted H2 2026, Tesla confirmed. August 2026: ~US$15bn raised, for customers who have not signed.

Why this is a neutrality problem, not a technology problem

A customer handing Intel a design hands it to a company that sells CPUs and is building AI accelerators. Intel's answer is a separate foundry subsidiary with its own P&L and board — but neutrality is demonstrated over years.

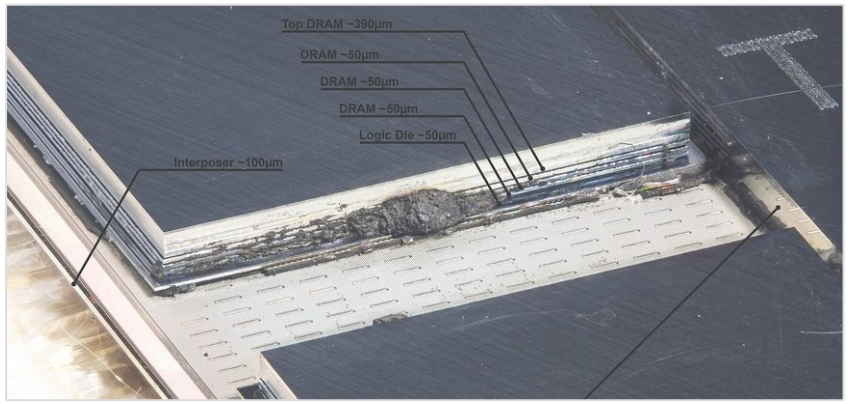
Track this: whether a named external customer tapes out a high-volume product. The rest is a press release.

Source: company announcements to August 2026; allocations per supply-chain reports.

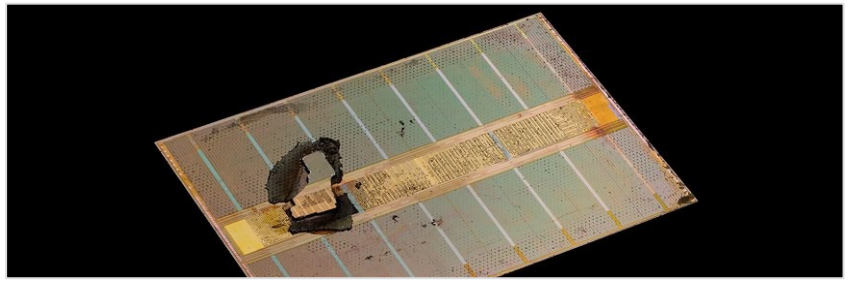
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Area 2 — advanced packaging: three approaches

When one die can no longer hold the design, how you join dies becomes the differentiating process.



2.5D package cross-section: logic die and HBM stacks on a shared interposer



An HBM DRAM die stack — the component that made packaging the bottleneck

TSMC — CoWoS · InFO · SoIC

- Large silicon interposer carries logic plus HBM (CoWoS-S), or an RDL/LSI variant (CoWoS-L).
- SoIC provides true 3D hybrid bonding, used in AMD's stacked cache and next-gen AI parts.
- Strength: volume and maturity. Weakness: interposer area is the constraint.

Intel — EMIB · EMIB-T · Foveros

- EMIB embeds a silicon bridge in the substrate — no interposer, so no reticle-area ceiling.
- EMIB-T (2026) routes power through the bridge, addressing HBM4 power delivery.
- Targeting 120×120mm packages against a 100×100mm norm. Malaysia online 2026.

Samsung — I-Cube · SAINT · H-Cube

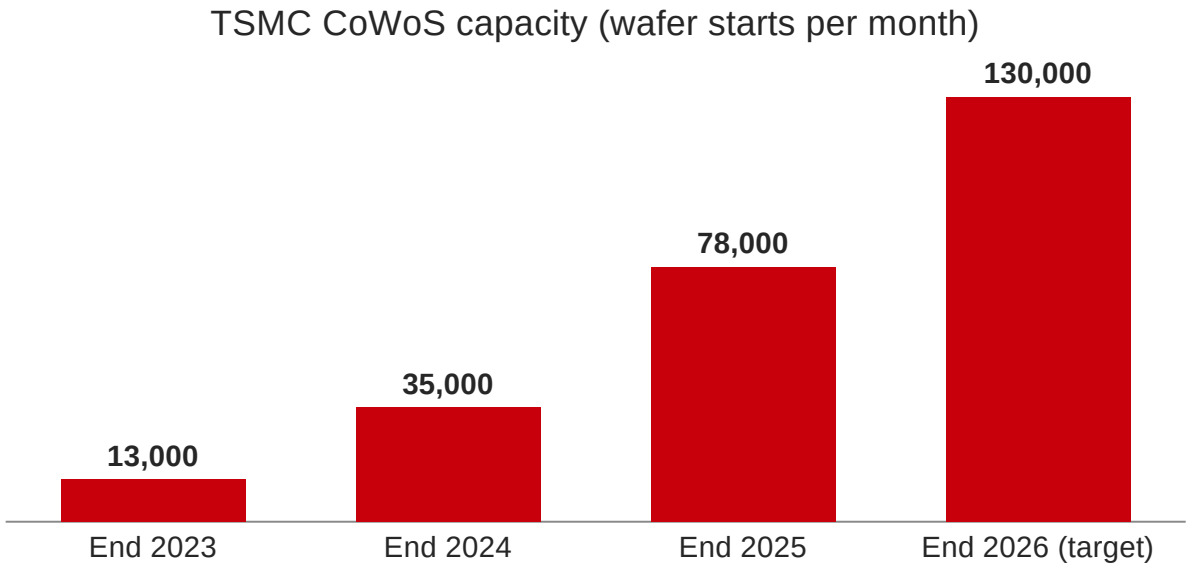
- I-Cube is the 2.5D analogue of CoWoS; SAINT covers 3D stacking.
- The pitch is a turnkey memory-plus-logic-plus-package offer nobody else can make.
- Same structural problem: it also sells HBM to your competitors.

Packaging was an OSAT commodity bought on price. It is now a foundry process, co-designed with the wafer.

Source: TrendForce and company packaging roadmaps, 2026; Intel newsroom. Photos: Wikimedia Commons.

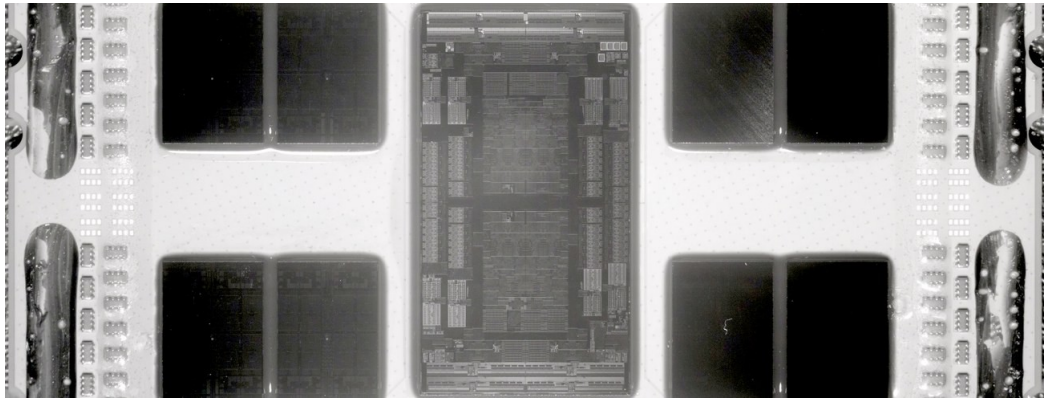
Area 2 continued — packaging capacity is the constraint

The constraint on AI silicon in 2026 is packaging capacity, not lithography.



Roughly a 10x expansion in three years — and still sold out

For two years the industry argued about nanometres while the allocation fight was over interposer area. To know who ships AI silicon in 2027, ask who holds packaging capacity.



Nine dies in one package — the package becomes the constraint

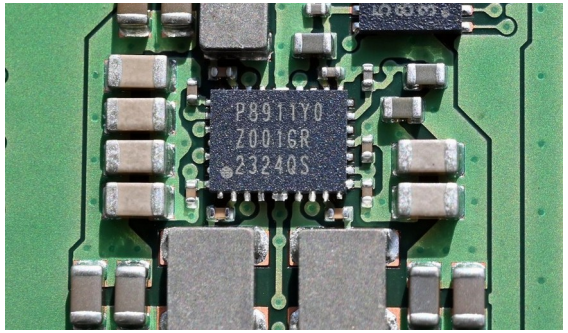
Supply–demand gap	~20% today, forecast to narrow to ~10% by end-2026
Booking status	CoWoS-L and CoWoS-S reported fully booked; 2nm booked into 2028
Where the overflow goes	ASE CoWoP, Amkor and SPIL are absorbing the tail — the first real crack in TSMC's packaging monopoly
Next platform	CoPoS (chip-on-panel-on-substrate) qualification running through 2026

Source: TrendForce (Jun 2026); TSMC Symposium 2026; capacity estimates. Approximate.

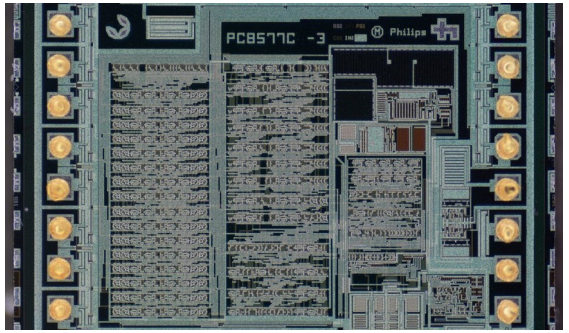
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Area 3 — mature nodes: price recovery over oversupply

2026 prices are rising. The structural oversupply that caused the 2023–25 slump has not gone away.



PMIC — where tightness sits



Display driver die

What is pushing prices up in 2026

- TSMC has cut 28nm output by over 25% since early 2026, moving capacity to advanced nodes and interposers. Spillover lands on UMC and Vanguard.
- UMC, VIS, PSMC and Nexchip all moved on price from around April 2026 — increases of up to 10%.
- Tightness sits in 8-inch PMIC, BCD, high-voltage, display driver, MCU, CIS and RF, plus some 12-inch 22/28nm and 40/55/65nm flows.

What has not changed underneath

- SMIC and Hua Hong keep expanding and discounting; China's build-out is policy-driven, not return-driven.
- UMC ran at 79% utilisation in Q1 2026, guiding to the low 80s in Q2 — a recovery, not a boom.
- UMC was reported cutting supplier prices 15% for 2026 — cost squeezed as price recovers.

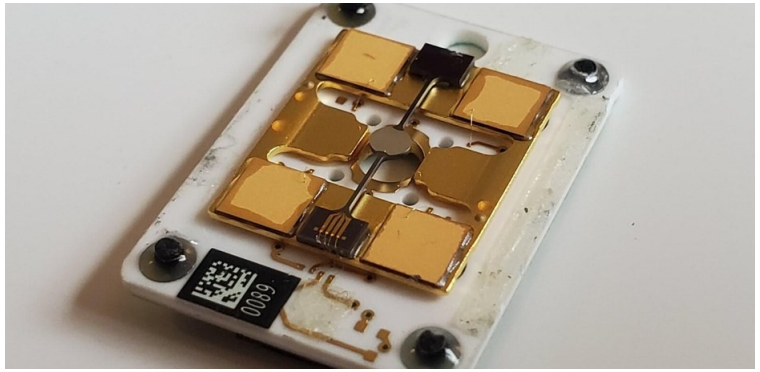
The 2026 price rise is supply-side, caused by TSMC reducing 28nm output, not by demand recovery. When Chinese 28nm capacity finishes ramping, the oversupply returns. Treat 2026 margin as a window, not a trend.

Source: TrendForce, Mar–Jun 2026; UMC Q1 2026 results. Photos: Wikimedia.

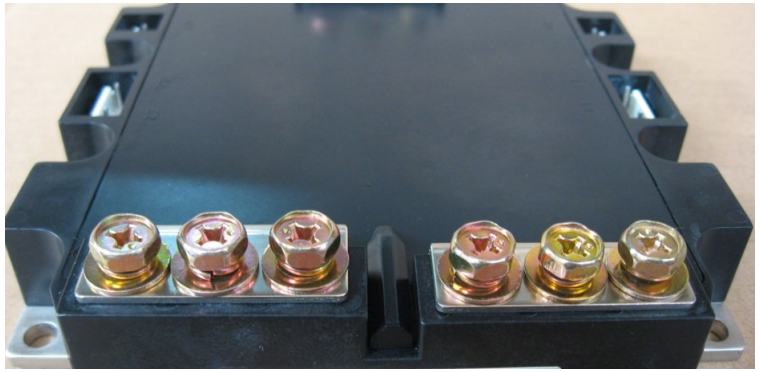
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Area 4 — specialty processes

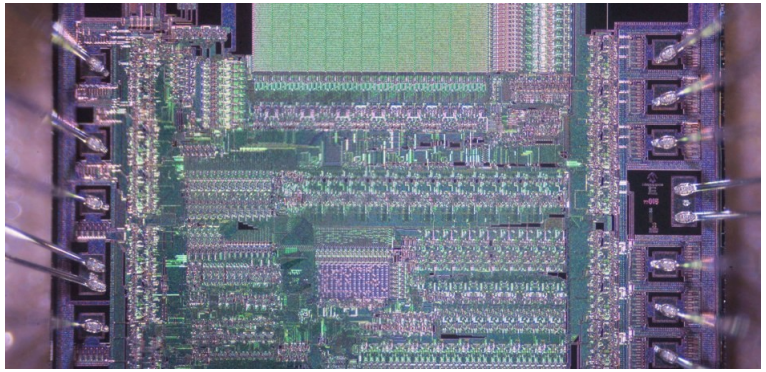
Competition here is on process recipes, qualification and long-term supply, not density.



A MEMS mirror die — every device its own process



Power modules — a decade-long qualification



Mixed-signal silicon — recipe, not density

PLAYER	PROCESS FRANCHISE	POSITION
GlobalFoundries	RF-SOI, FDX, automotive	The company that quit 7nm and became profitable doing it
Tower	Analog, RF, power, CIS	The Intel acquisition failed on China approval; still independent
X-FAB	MEMS, SiC, automotive analog	Europe's specialty pure-play; every process is bespoke
Vanguard (VIS)	Power, display driver, 8-inch	Building 12-inch with NXP in Singapore
DB HiTek	BCD, power, RF	Korea's specialty foundry, consistently profitable at 0.5% share
SkyWater	US Trusted Foundry, MEMS	Defence and R&D niches; open-source PDK for 130nm

Why it works: a qualified automotive process must be supplied for a decade.

Why it is not growth: the market grows with cars, not AI.

Source: Company disclosures, 2026. Photos: Wikimedia Commons.

Area 5 — geopolitics and subsidy

Three overseas fabs, and what capacity costs away from its established ecosystem.



TSMC Arizona (Fab 21)

Phase 1 in volume at 4nm, ~24k wpm. Phase 2 tool-in Q3 2026 for 3nm/2nm in 2027. Phase 3 targets 2nm and A16.



TSMC Kumamoto (JASM)

Fab 1 in volume since Q4 2024 at 28/22 and 16/12nm, ~55k wpm. Fab 2 (6/7nm) targeting 2027.



TSMC Dresden (ESMC)

28/22 and 16/12nm, ~40k wpm planned. Started Q3 2024, production H2 2027 — the slowest of the three.

Section 232 tariffs

25% on certain imported semiconductors from 15 Jan 2026, scoped by tensor performance and DRAM bandwidth. A second phase is expected.

The CHIPS tax credit

The 25% investment credit rose to 35% — and expires end-2026. That date is doing a lot of work in current capex plans.

Export controls

No EUV for China. SMIC's 7nm is DUV multi-patterning: it works, at lower yield and higher cost, and stops short of 3nm.

Overseas fabs cost more per wafer and reach yield later: diversification is bought with margin. So far the customer pays.

Source: Section 232 proclamation (Jan 2026); ITIF (Jun 2026); TSMC disclosures; NIST CHIPS.

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Area 6 — ecosystem lock-in

Why a better transistor alone does not move a customer.

EDA certification

Synopsys and Cadence flows are certified per foundry, per node. TSMC's Open Innovation Platform has thirty years behind it.

Silicon-proven IP

Thousands of hard IP blocks exist on TSMC nodes and often nowhere else. Porting one is a project, not a purchase.

PDK maturity

Models correlated against millions of wafers. A young PDK forces added margin — lost performance in the product.

Shuttle cadence

Multi-project runs let a customer test before committing to a mask set. Thin calendars are a real barrier.

Design services

A network of design houses fluent in one foundry's flow. Capacity to execute is part of the offer.

Reference designs

Chiplet standards, HBM interface kits, packaging co-design — what turns a node into a product.

The loop, stated once and for all

Neutrality attracts customers → **Customers bring volume** → **Volume buys yield learning** → **Yield buys margin** → **Margin funds the next node** → **The next node attracts customers**

A challenger must break the loop somewhere. Intel tries at neutrality, Samsung at margin, China at volume. None has closed it.

Source: Foundry ecosystem documentation; author's synthesis.

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Sixteen foundries compared

The same fields for every company, so different strategies can be compared.

COMPANY	MODEL	MOST ADVANCED NODE IN HVM	Q1'26 SHARE	PRIMARY DEMAND	STRUCTURAL WEAKNESS
TSMC	Pure-play	**N2 (2nm)**	**72.3%**	AI accelerators, mobile SoC, HPC, all custom silicon	Geographic concentration; customers who would fund an alternative
Samsung Foundry	IDM foundry	**SF2 (2nm)**	**6.5%**	Captive Exynos, some mobile and crypto ASIC	Competes with its own customers; yield reputation
SMIC	National champ.	**7nm (DUV)**	**5.1%**	Huawei, domestic AI and mobile, mature mixed-signal	No EUV; yield and cost penalty; cannot scale below ~5nm
UMC	Pure-play	**14nm**	**3.9%**	Display driver, MCU, connectivity, RF, auto	Stopped at 14nm; caught between TSMC above and China below
GlobalFoundries	Pure-play	**12nm FDX**	**3.3%**	RF-SOI, automotive, IoT, US/EU secure supply	No leading-edge path; growth tied to a slow-growing end market
Hua Hong Group	National champ.	**7nm (ramping)**	**2.5%**	Power, analog, eNVM, CIS, MCU	Same equipment ceiling as SMIC; policy-driven capex
Intel Foundry	IDM foundry	**18A**	**n/a***	Intel's own products; Tesla on 14A	External customers not yet at scale; capacity ahead of demand
Tower	Specialty	**65nm**	**0.8%**	Analog, RF, power, CIS, silicon photonics	Sub-scale; failed Intel acquisition left it standalone
Nexchip	National champ.	**28nm**	**0.8%**	Display driver, CIS, mature logic	Price-led entry; thin margin, heavy domestic competition
VIS (Vanguard)	Specialty	**0.11µm**	**0.8%**	Power, display driver, 8-inch mixed signal	8-inch heavy; 12-inch build with NXP is a bet on capacity timing
PSMC	Pure-play	**28nm**	**0.8%**	Memory foundry, CIS, power, MCU	Memory exposure adds cycle volatility to a foundry P&L
X-FAB	Specialty	**130nm**	**<0.5%**	MEMS, SiC, automotive analog, medical	Very small scale; depends on European industrial demand
DB HiTek	Specialty	**90nm**	**<0.5%**	BCD power, RF, display	Narrow franchise; limited capacity to expand
SkyWater	Specialty	**90nm**	**<0.5%**	US Trusted Foundry, defence, MEMS, R&D	Revenue depends on government programmes
Rapidus	National champ.	**2nm (pilot)**	**—**	Target: Japanese and US AI/edge customers	No production revenue; HVM target 2027; unproven at scale
Tata Electronics	National champ.	**28nm (build)**	**—**	Target: automotive, display, domestic Indian demand	First fab not yet producing; ecosystem being built from zero

* Intel Foundry external revenue is not ranked in the TrendForce top-10 pure-play table.

Full one-page cards are in the appendix.

Source: TrendForce Q1 2026; company disclosures. Shares are of top-10 foundry revenue.

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Four things to check over the next 18 months

Observable events with dates, not predictions.

01	**Does Intel land a named external 14A customer at volume?** Watch for: a high-volume tape-out announced by the customer, not by Intel. First 14A commitments guided H2 2026.	If yes — the leading edge is a three-way market for the first time since 2015. If no — the US$15bn raise looks stranded.
02	**Does CoWoS supply actually catch demand?** Watch for: TSMC's packaging capacity against the 120–140k wpm end-2026 target, and whether the gap narrows to ~10%.	If it closes — accelerator pricing loosens and OSAT overflow shrinks. If not — packaging stays the lever.
03	**When does the mature-node overhang return?** Watch for: UMC and VIS utilisation, and the ramp of Chinese 28nm. The 2026 rise is supply-side.	If Chinese capacity lands as planned — the rebound reverses in 2027. If domestic demand absorbs it — the overhang is smaller.
04	**Do Rapidus and India become real?** Watch for: Rapidus HVM in H2 2027 with a named customer; Tata's first fab producing qualified parts.	If either works — the national-champion model is validated outside China and the mid nodes get crowded.

Each of these resolves within 18 months.

Source: Company guidance and press reporting as of August 2026.

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What this means for us

Three supply-chain exposures worth tracking, and where the leverage actually sits.

Single points of failure

- Leading-edge logic and advanced packaging are effectively one supplier, in one geography.
- The second source that matters is packaging capacity, not wafer starts — that is where OSATs are taking share.
- Any AI roadmap dependency should be stated as a CoWoS-class allocation risk, not a node risk.

Cost direction

- Wafer cost per node is rising faster than transistor cost is falling; the next node is no longer cheaper per function.
- Section 232 tariffs and overseas-fab premiums both land on the bill of materials.
- Mature-node relief in 2026 is a window that may close in 2027.

Where to watch, concretely

- TSMC monthly revenue and quarterly capex splits — the earliest public signal of allocation shifts.
- TrendForce quarterly foundry ranking, but check the denominator each time.
- Any named external tape-out at Intel 14A or Samsung SF2 — the only real evidence of a second source.

These numbers have a shelf life of about a quarter. The appendix carries a source-and-capture-date page: refreshing means updating one table.

Source: Author's synthesis.

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Summary

01 **The barrier is capital, not physics.**

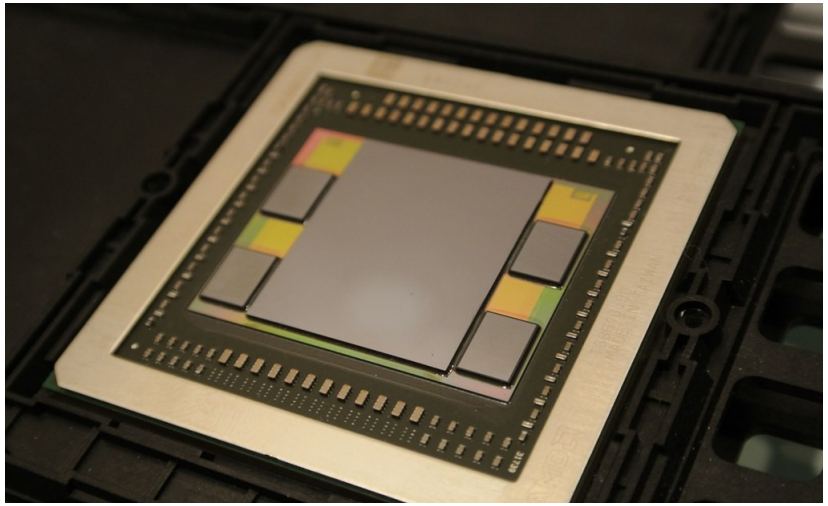
Twenty-three companies left the leading edge without losing an engineering argument. They ran out of volume to amortise the next node.

02 **The barrier is switching cost, not process technology.**

PDKs, silicon-proven IP, EDA certification and requalification keep a customer in place when a rival offers a better transistor.

03 **The next constraint is packaging.**

For two years the industry argued about nanometres while the binding constraint was interposer area. Watch the back end.



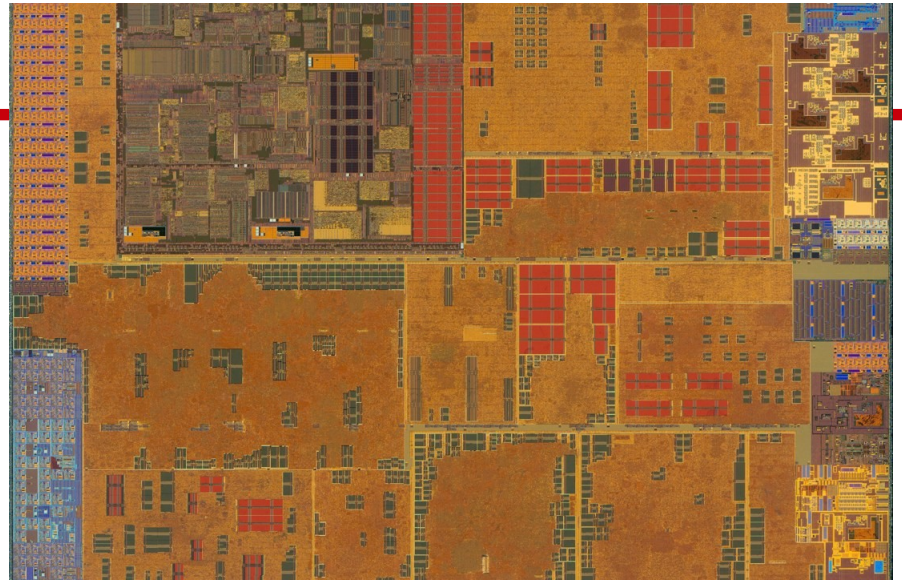
Back to page 2. A dozen companies touch this package. One of them can manufacture it.

That is the industry.

Source: Photo: Wikimedia Commons.

Appendix

Glossary, data sources and refresh schedule



FUTUREWEI
Technologies

Glossary

Terms used in this deck that were not covered in the fabrication deck.

GAA / nanosheet	Gate-all-around. The gate wraps the channel on all sides, replacing the FinFET's three.
RibbonFET	Intel's branding for its nanosheet GAA implementation.
CoWoS	Chip-on-Wafer-on-Substrate. TSMC's 2.5D platform; logic and HBM share a silicon interposer.
EMIB / EMIB-T	Intel's embedded silicon bridge; the -T variant routes power through the bridge.
SAINT / I-Cube	Samsung's 3D and 2.5D packaging platforms.
PDK	Process design kit. The models, rules and libraries a foundry supplies to designers.
D0	Defect density — the underlying metric behind yield at a given die size.
RF-SOI / FDX	RF silicon-on-insulator and fully-depleted SOI — GlobalFoundries' specialty franchises.
Foundry 2.0	TSMC's 2024 market definition adding packaging, test, masks and non-memory IDM output.
MBCFET	Samsung's branding for its nanosheet GAA implementation.
BSPDN / PowerVia	Power rails move to the back of the wafer, freeing front-side metal for signal routing.
SoIC	TSMC's 3D stacking platform using hybrid bonding, no microbumps.
Foveros	Intel's 3D die-stacking platform.
CoPoS	Chip-on-Panel-on-Substrate. Panel-level successor to CoWoS, in qualification.
MPW / shuttle	Multi-project wafer. Many designs share one mask set to cut prototyping cost.
BCD	Bipolar-CMOS-DMOS. The standard process family for power management ICs.
eNVM	Embedded non-volatile memory — flash or MRAM integrated into a logic process.
Section 232	US trade statute used for the January 2026 semiconductor import tariffs.

Source: Definitions per company technical documentation.

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Data sources and refresh schedule

Process physics does not expire; share, capacity and pricing do. Time-sensitive figures are listed here.

PAGE	FIGURE	SOURCE	REFRESH
20	Foundry revenue share, Q1 2026	TrendForce top-10 ranking, published Jun 2026	**Quarterly**
21	Total foundry market $320bn (2025); Foundry 2.0 definition	Tom's Hardware / TrendForce; TSMC (Jul 2024)	**Annual**
12, 22	Wafer prices by node; design cost by node	Silicon Analysts 2026 series; IBS estimates	**Semi-annual**
23	TSMC capex 2026 US$52–64bn	TrendForce and company guidance, revised twice in 2026	**Quarterly**
24	Utilisation: UMC 79% Q1'26; SMIC 28nm >95%; N2 yield ~65%	Company earnings calls; supply-chain reports	**Quarterly**
27, 28	2nm node status; Intel 14A customer commitments	Company announcements through Aug 2026	**Continuous**
30	CoWoS capacity 13k→130k wpm; 20%→10% supply gap	TrendForce Jun 2026; TSMC symposium May 2026	**Quarterly**
31	Mature-node price moves; TSMC 28nm output cut >25%	TrendForce Mar–Jun 2026	**Quarterly**
33	Section 232 25% tariff; CHIPS ITC 35% expiring end-2026; overseas fab schedules	US proclamation Jan 2026; ITIF Jun 2026; NIST; TSMC	**Event-driven**
35	Sixteen-player comparison matrix	Aggregates all of the above	**Quarterly**

Captured 25 August 2026. Where a figure is an estimate — yields, utilisation, wafer prices, packaging capacity — the slide says so. Refresh this table first.

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People

Morris Chang (; MIT 1951 yearbook; APEC 2021) · Carver Mead (CHM, 2005) · Lynn Conway (2006) · Robert Noyce and Gordon Moore (Intel, 1970) · Jerry Sanders (AMD)

Fabs and facilities

TSMC Fab 21 Arizona · JASM Kumamoto · ESMC Dresden · GlobalFoundries Dresden (ex-AMD Fab 36) · UMC Hsinchu · SMIC Shenzhen · Hsinchu Science Park · Powerchip · X-FAB Erfurt · wafer-track line and photolithography cleanroom (Wafertraksystem)

Devices and materials

AMD Fiji package · AMD Epyc 7702 · Xilinx XC4010 FPGA die · Qualcomm QSD8250 · NVIDIA Turing dies · Broadcom BCM2711 · Intel 8086, 8088, CE3100 · TI CC2530 · PIC16C57 · NXP PCF8577C · Hynix DDR5 PMIC · 14.5 MP CMOS sensor · MEMS gyroscope · MEMS mirror · Mitsubishi IGBT · SiC ingot · germanium ICs · FinFET/nanosheet · 2.5D cross-section · HBM stack

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